Probability and Statistics

Joint Probability Distributions and Random Samples

(PART 1)

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1 Jointly Distributed Random Variables

2 Expected Values, Covariance, and Correlation

3 Statistics and Their Distributions

Jointly Distributed Random Variables

- Joint probability distribution（联合概率分布）

Let X and Y be two discrete rv's defined on the sample space $\mathcal{S}$ of an experiment. The **joint probability mass function** $p(x, y)$ is defined for each pair of numbers (x, y) by

$$p(x, y) = P(X = x \text{ and } Y = y)$$

It must be the case that $p(x, y) \geq 0$ and $\sum_x \sum_y p(x, y) = 1$.

Now let A be any set consisting of pairs of (x, y) values (e.g., $A = \{(x, y): x + y = 5\}$ or $\{(x, y): \max(x, y) \leq 3\}$). Then the probability $P[(X, Y) \in A]$ is obtained by summing the joint pmf over pairs in A:

$$P[(X, Y) \in A] = \sum_{(x, y) \in A} \sum p(x, y)$$

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Jointly Distributed Random Variables

- Joint probability distribution（联合概率分布）

- If we have two discrete random variables X and Y and their possible values are 100, 250 and 0, 100, 200, respectively, with the following joint probability

$p(x, y)$		y = 0	100	200
x	100	.20	.10	.20
	250	.05	.15	.30

- The probability that Y is larger than 100 is

$$P(Y \geq 100) = p(100, 100) + p(250, 100) + p(100, 200) + p(250, 200)$$
$$= .75$$

Jointly Distributed Random Variables

- Marginal probability distribution (边缘概率分布)

The **marginal probability mass functions** of X and of Y, denoted by $p_X(x)$ and $p_Y(y)$, respectively, are given by

$$p_X(x) = \sum_y p(x, y) \qquad p_Y(y) = \sum_x p(x, y)$$

Jointly Distributed Random Variables

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$$p_X(x) = \sum_y p(x, y) \qquad p_Y(y) = \sum_x p(x, y)$$

$p(x, y)$		y = 0	100	200
x	100	.20	.10	.20
	250	.05	.15	.30

$$p_X(100) = p(100, 0) + p(100, 100) + p(100, 200) = .50$$

and

$$p_X(250) = p(250, 0) + p(250, 100) + p(250, 200) = .50$$

The marginal pmf of X is then

$$p_X(x) = \begin{cases} .5 & x = 100, 250 \\ 0 & \text{otherwise} \end{cases}$$

Jointly Distributed Random Variables

- Joint probability distribution of continuous random variables

Let X and Y be continuous rv's. A **joint probability density function** $f(x, y)$ for these two variables is a function satisfying $f(x, y) \geq 0$ and $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x, y)\, dx\, dy = 1$. Then for any two-dimensional set A

$$P[(X, Y) \in A] = \int_A \int f(x, y)\, dx\, dy$$

In particular, if A is the two-dimensional rectangle $\{(x, y): a \leq x \leq b, c \leq y \leq d\}$, then

$$P[(X, Y) \in A] = P(a \leq X \leq b, c \leq Y \leq d) = \int_a^b \int_c^d f(x, y)\, dy\, dx$$

Jointly Distributed Random Variables

- Marginal probability distribution of continuous random variables

The **marginal probability density functions** of X and Y, denoted by $f_X(x)$ and $f_Y(y)$, respectively, are given by

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y)\, dy \qquad \text{for } -\infty < x < \infty$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y)\, dx \qquad \text{for } -\infty < y < \infty$$

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Jointly Distributed Random Variables

- Marginal probability distribution of continuous random variables
- If we have two continuous random variables X and Y with the following joint probability distribution

$$f(x, y) = \begin{cases} \frac{6}{5}(x + y^2) & 0 \le x \le 1, 0 \le y \le 1 \\ 0 & \text{otherwise} \end{cases}$$

- The marginal probability distribution of X is

Jointly Distributed Random Variables

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$$f(x, y) = \begin{cases} \frac{6}{5}(x + y^2) & 0 \le x \le 1, 0 \le y \le 1 \\ 0 & \text{otherwise} \end{cases}$$

- The marginal probability distribution of X is

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y)\, dy = \int_0^1 \frac{6}{5}(x + y^2)\, dy = \frac{6}{5}x + \frac{2}{5}$$

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Jointly Distributed Random Variables

- Joint probability distribution of more than two random variables

If $X_1, X_2, \ldots, X_n$ are all discrete random variables, the joint pmf of the variables is the function

$$p(x_1, x_2, \ldots, x_n) = P(X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n)$$

If the variables are continuous, the joint pdf of $X_1, \ldots, X_n$ is the function $f(x_1, x_2, \ldots, x_n)$ such that for any n intervals $[a_1, b_1], \ldots, [a_n, b_n]$,

$$P(a_1 \le X_1 \le b_1, \ldots, a_n \le X_n \le b_n) = \int_{a_1}^{b_1} \cdots \int_{a_n}^{b_n} f(x_1, \ldots, x_n)\, dx_n \cdots dx_1$$

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Jointly Distributed Random Variables

- Conditional probability distribution

Let X and Y be two continuous rv's with joint pdf $f(x, y)$ and marginal X pdf $f_X(x)$. Then for any X value x for which $f_X(x) > 0$, the **conditional probability density function of Y given that $X = x$** is

$$f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)} \qquad -\infty < y < \infty$$

If X and Y are discrete, replacing pdf's by pmf's in this definition gives the **conditional probability mass function of Y when $X = x$.**

1 Jointly Distributed Random Variables

2 Expected Values, Covariance, and Correlation

3 Statistics and Their Distributions

Expected Values, Covariance, and Correlation

- Expected value of function of two random variables

Let X and Y be jointly distributed rv's with pmf $p(x, y)$ or pdf $f(x, y)$ according to whether the variables are discrete or continuous. Then the expected value of a function $h(X, Y)$, denoted by $E[h(X, Y)]$ or $\mu_{h(X, Y)}$, is given by

$$E[h(X,Y)] = \begin{cases} \sum_{x} \sum_{y} h(x, y) \cdot p(x, y) & \text{if } X \text{ and } Y \text{ are discrete} \\ \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x, y) \cdot f(x, y)\, dx\, dy & \text{if } X \text{ and } Y \text{ are continuous} \end{cases}$$

Expected Values, Covariance, and Correlation

- Covariance (协方差)

The **covariance** between two rv's X and Y is

$$\mathrm{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

$$= \begin{cases} \sum_{x}\sum_{y} (x - \mu_X)(y - \mu_Y)p(x, y) & X, Y \text{ discrete} \\ \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} (x - \mu_X)(y - \mu_Y)f(x, y)\, dx\, dy & X, Y \text{ continuous} \end{cases}$$

$$\mathrm{Cov}(X, Y) = E(XY) - \mu_X \cdot \mu_Y$$

- Strong positive relationship
  - Either large X occurs with large Y or small X with small Y, so $Cov(X, Y) > 0$
- Strong negative relationship
  - Either large X occurs with small Y or small X withlarge Y, so $Cov(X, Y) < 0$

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Expected Values, Covariance, and Correlation

- Correlation coefficient (相关系数)

The **correlation coefficient** of X and Y, denoted by $\mathrm{Corr}(X, Y)$, $\rho_{X,Y}$, or just ρ, is defined by

$$\rho_{X,Y} = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \cdot \sigma_Y}$$

1. If a and c are either both positive or both negative,

$$\mathrm{Corr}(aX + b, cY + d) = \mathrm{Corr}(X, Y)$$

2. For any two rv's X and Y, $-1 \le \mathrm{Corr}(X, Y) \le 1$.

Expected Values, Covariance, and Correlation

- Correlation coefficient (相关系数)

1. If a and c are either both positive or both negative,

$$\text{Corr}(aX + b, cY + d) = \text{Corr}(X, Y)$$

2. For any two rv's X and Y, $-1 \leq \text{Corr}(X, Y) \leq 1$.

1. **Covariance Calculation**:

$$\text{Cov}(X', Y') = \text{Cov}(aX + b, cY + d).$$

Using the linearity property of covariance, we get:

$$\text{Cov}(X', Y') = \text{Cov}(aX, cY) = ac\,\text{Cov}(X, Y).$$

Here, the constants b and d do not affect the covariance since they are just shifts (constants).

2. **Standard Deviation Calculation**:
   - The standard deviation of X' is:

$$\sigma_{X'} = \sqrt{\text{Var}(X')} = \sqrt{\text{Var}(aX + b)} = |a|\sigma_X.$$

   - Similarly, the standard deviation of Y' is:

$$\sigma_{Y'} = \sqrt{\text{Var}(Y')} = \sqrt{\text{Var}(cY + d)} = |c|\sigma_Y.$$

3. **Correlation Calculation**: The correlation coefficient of X' and Y' is:

$$\text{Corr}(X', Y') = \frac{\text{Cov}(X', Y')}{\sigma_{X'}\sigma_{Y'}} = \frac{ac\,\text{Cov}(X, Y)}{|a|\sigma_X\,|c|\sigma_Y}.$$

Since a and c are either both positive or both negative, ac has the same sign as $|a||c|$, so:

$$\text{Corr}(X', Y') = \frac{\text{Cov}(X, Y)}{\sigma_X\sigma_Y} = \text{Corr}(X, Y).$$

For any two random variables X and Y, the **Cauchy-Schwarz inequality** states:

$$(\mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])])^2 \leq \mathbb{E}[(X - \mathbb{E}[X])^2] \cdot \mathbb{E}[(Y - \mathbb{E}[Y])^2].$$

Let's express this inequality in terms of **covariance** and **variance**:

- The **covariance** of X and Y is defined as:

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])].$$

- The **variance** of X is:

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \sigma_X^2.$$

- The **variance** of Y is:

$$\text{Var}(Y) = \mathbb{E}[(Y - \mathbb{E}[Y])^2] = \sigma_Y^2.$$

Substituting these definitions, the Cauchy-Schwarz inequality becomes:

$$(\text{Cov}(X, Y))^2 \leq \text{Var}(X) \cdot \text{Var}(Y).$$

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Expected Values, Covariance, and Correlation

- Correlation coefficient (相关系数)
- Correlation coefficient measures linear relationship between X and Y

1. If X and Y are independent, then $\rho = 0$, but $\rho = 0$ does not imply independence.
2. $\rho = 1$ or -1 iff $Y = aX + b$ for some numbers a and b with $a \neq 0$.

Independence implies $\rho = 0$:

- If X and Y are independent, then $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$.
- The definition of the correlation coefficient is:

$$\rho = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y},$$

where $\mathrm{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$.

- If X and Y are independent, then:

$$\mathrm{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 0.$$

Thus, $\rho = 0$.

$\rho = 0$ does not imply independence:

- A correlation coefficient of 0 indicates that X and Y are uncorrelated, but this does not imply independence. Uncorrelated variables can still have a nonlinear relationship.
- **Example**: Let X be a standard normal random variable, and let $Y = X^2$. Then:

$$\mathbb{E}[XY] = \mathbb{E}[X \cdot X^2] = \mathbb{E}[X^3] = 0$$

(since the odd moments of a standard normal variable are 0). Thus:

$$\mathrm{Cov}(X,Y) = 0,$$

but X and Y are clearly dependent because Y is determined by X.

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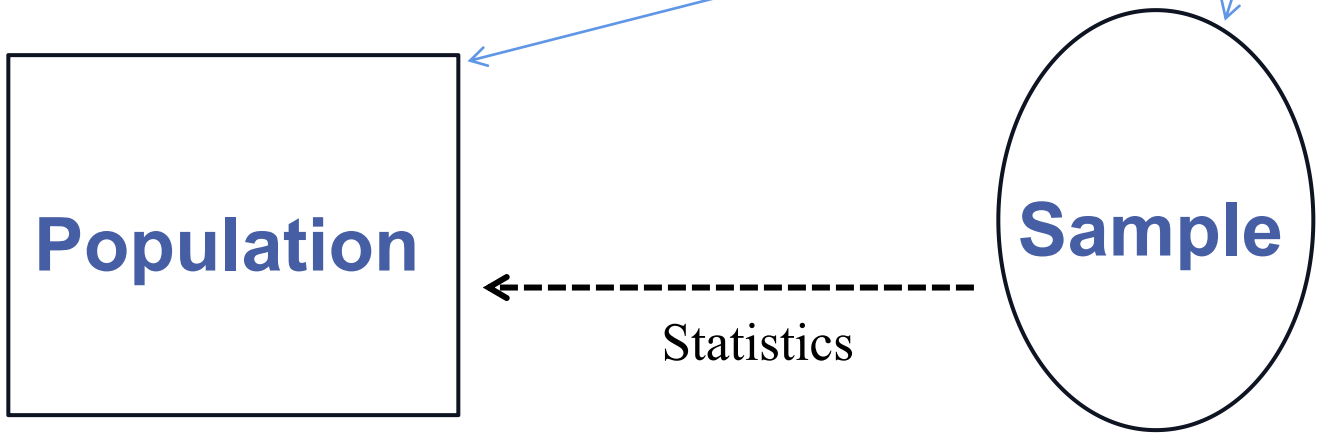
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3 Statistics and Their Distributions

Population and sample

- The relationship between statistics and probability
  - There are 3 male students among 30 randomly selected students
  - How many male students can be there in the entire class?



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Statistics and Their Distributions

- Statistic (统计量)

A **statistic** is any quantity whose value can be calculated from sample data. Prior to obtaining data, there is uncertainty as to what value of any particular statistic will result. Therefore, a statistic is a random variable and will be denoted by an uppercase letter; a lowercase letter is used to represent the calculated or observed value of the statistic.

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Statistics and Their Distributions

- Statistic (统计量)
- Suppose the length of a material follows a Weibull distribution with parameters $\alpha = 2, \beta = 5$, and we generate six different samples, each with 10 data. The following table shows generated sample data and corresponding sample mean $\bar{x}$, sample median $\tilde{x}$, and sample standard deviation s
- Those $\bar{X}$, $\tilde{X}$ and S are statistics and themselves are random variables on its own

Sample	1	2	3	4	5	6
1	6.1171	5.07611	3.46710	1.55601	3.12372	8.93795
2	4.1600	6.79279	2.71938	4.56941	6.09685	3.92487
3	3.1950	4.43259	5.88129	4.79870	3.41181	8.76202
4	0.6694	8.55752	5.14915	2.49759	1.65409	7.05569
5	1.8552	6.82487	4.99635	2.33267	2.29512	2.30932
6	5.2316	7.39958	5.86887	4.01295	2.12583	5.94195
7	2.7609	2.14755	6.05918	9.08845	3.20938	6.74166
8	10.2185	8.50628	1.80119	3.25728	3.23209	1.75468
9	5.2438	5.49510	4.21994	3.70132	6.84426	4.91827
10	4.5590	4.04525	2.12934	5.50134	4.20694	7.26081
$\bar{x}$	4.401	5.928	4.229	4.132	3.620	5.761
$\tilde{x}$	4.360	6.144	4.608	3.857	3.221	6.342
s	2.642	2.062	1.611	2.124	1.678	2.496

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Statistics and Their Distributions

- Sampling distribution （抽样分布）
- Any statistic, being a random variable, has a probability distribution. In particular, the sample mean $\overline{X}$ has a probability distribution
- The probability distribution of a statistic is referred to as its sampling distribution

Statistics and Their Distributions

- Random sampling （随机抽样）
- The probability distribution of any particular statistic depends on: population distribution (normal, uniform, etc.), the sample size *n,* and the method of sampling

- The most widely used sampling method is random sampling:



Statistics and Their Distributions

- Random sampling （随机抽样）
- The probability distribution of any particular statistic depends on: population distribution (normal, uniform, etc.), the sample size *n,* and the method of sampling

- The most widely used sampling method is random sampling:

> The rv's $X_1, X_2, \ldots, X_n$ are said to form a (simple) **random sample** of size n if
>
> **1.** The X_is are independent rv's.
>
> **2.** Every X_i has the same probability distribution.

- If sampling is either with replacement or from an infinite population, those two conditions can be satisfied and this sampling method is the random sampling

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Statistics and Their Distributions

- There are two general methods for obtaining information about a statistic's sampling distribution: calculation and simulation
- A discrete random variable X has the following PMF

x	40	45	50
$p(x)$	.2	.3	.5

with $\mu = 46.5, \sigma^2 = 15.25$

- Now, we want to know the probability distribution of sample mean and sample variance with sample size of 2
- For example, $\bar{x} = 45$ occurs three times in the table with probabilities 0.10, 0.09 and 0.10, so

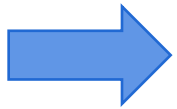
$$p_{\bar{X}}(45) = P(\bar{X} = 45) = .10 + .09 + .10 = .29$$

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Statistics and Their Distributions

- Calculate one by one in the same way we have

x_1	x_2	$p(x_1, x_2)$	$\bar{x}$	s^2
40	40	.04	40	0
40	45	.06	42.5	12.5
40	50	.10	45	50
45	40	.06	42.5	12.5
45	45	.09	45	0
45	50	.15	47.5	12.5
50	40	.10	45	50
50	45	.15	47.5	12.5
50	50	.25	50	0



$\bar{x}$	40	42.5	45	47.5	50
$p_{\bar{X}}(\bar{x})$	.04	.12	.29	.30	.25
s^2	0	12.5	50		
$p_{S^2}(s^2)$	.38	.42	.20		

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